

# Supporting Information

## Topological Systems with CP2K

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### S1 Scope and Reproduction Boundaries

This working document combines the retained band-topology calculations transferred from the k-point manuscript with separately validated transport, localizer, and quadratic-pseudospectrum developments. The original results are not recomputed or relabelled by moving the manuscript text. The development status is recorded as of 24 September 2026. A source freeze and complete public reproduction archive remain submission requirements.

#### Applications versus test calculations

TopoHSE-DB and the submitted generative-predictive discovery study are cited in the main text as materials-science applications. Their VASP-based calculations are not part of the CP2K validation records collected here. In particular, their database labels and candidate counts do not validate the new CP2K localizer or transport implementations.

The smaller calculations below are tests and demonstrations. The companion k-point paper additionally reports 18 converged Si/Al calculations in conventional cubic cells with eight Si atoms ( $a = 5.43$  Å) and four Al atoms ( $a = 4.05$  Å). These use PBE, DZVP-MOLOPT-SR-GTH bases, GTH-PBE pseudopotentials, 600/60 Ry grid cutoffs, and an SCF threshold of  $10^{-9}$ . Al uses Fermi-Dirac occupations at 1000 K, so its compared quantity is the electronic free energy. The same-mesh comparison uses  $4 \times 4 \times 4$  sampling: 64 full-grid points, 32 with time reversal, and four with either K290 or SPGLIB atomic symmetry. The maximum cell-energy/free-energy difference is  $1.4 \times 10^{-12}$  hartree; the Al atomic-symmetry stress difference is 0.098 bar. These tests demonstrate representation consistency, not independently converged material properties or general force accuracy. Their source and raw records remain those of the companion manuscript; no new Si/Al calculations were performed for the present discussion.

### S2 Native Wilson-Loop Test and Demonstration Calculations

The topology examples form a separate calculation set from the Si/Al mainline tests. The native extension supplies directed Gaussian-basis overlaps, second-variational spinors, and the Wilson-loop analysis described in the main text. The retained calculations were completed on 17 September 2026 with one MPI process and one OpenMP thread. Source provenance and artifact checksums are recorded in `validation/topology/README.md` and the figure manifest.

Both examples use PBE, the DZVP-MOLOPT-GGA-GTH basis from `BASIS_MOLOPT_UZH`, SOC pseudopotentials from `GTH_SOC_POTENTIALS`, a 200 Ry density cutoff and a 40 Ry relative cutoff. The potential labels are `GTH-PBE-q8` for neon and `GTH-PBE-q4` for tin. Complex orbitals and explicit full SCF meshes are used. The neon reference contains one atom at fractional coordinate  $(1/2, 1/2, 1/2)$  in a cubic cell of side 5 Å, with a  $2 \times 2 \times 2$  SCF mesh and convergence threshold  $10^{-9}$ . The stanene cell has lattice vectors  $(4.674, 0, 0)$ ,  $(2.337, 4.047802737288466, 0)$ , and  $(0, 0, 20)$  Å, and two tin atoms at fractional positions  $(1/3, 1/3, 0.5213)$  and  $(2/3, 2/3, 0.4787)$ . It therefore

has a buckling height of 0.852 Å. These fixed geometric parameters follow the pristine-stanene reference reported by Zhang et al.<sup>1</sup> An  $8 \times 8 \times 1$  SCF mesh and convergence threshold  $10^{-8}$  are used for stanene. The examples test the topology implementation at these fixed settings, rather than constitute a new structural or band-gap prediction.

Eight occupied spinors are retained in each case. The stanene second-variational calculation includes all 26 scalar basis states, giving 52 spinor states before occupied-band selection. For neon, five scalar states produce ten spinors. The selected half-plane is  $\mathbf{k}(s, t) = (t, s/2, 0)$  in fractional reciprocal coordinates, with  $0 \leq s, t \leq 1$ . The loop closes with reciprocal shift  $(1, 0, 0)$ . Starting meshes contain four points on each of three loops for neon and 12 points on each of 13 loops for stanene. Refinement doubles the points per loop and inserts a midpoint between neighboring transverse loops. The accepted surfaces have  $8 \times 5$  and  $192 \times 193$  points, respectively. The last center changes are  $1.28 \times 10^{-5}$  and  $2.73 \times 10^{-4}$  in reduced units. The smallest link singular values are 0.99826 and 0.95248.

The native invariants are zero for neon and one for stanene. Z2Pack 2.2.1 gives the same stanene parity using either the raw overlaps or their unitary polar factors. For the latter, the maximum Wilson-eigenvalue difference from the native calculation is  $7.95 \times 10^{-14}$ . This comparison tests the loop algebra with identical input matrices. Separate operator checks enforce zero-step normalization, reverse-link adjoint relations, loop reversal, and atomic-position phases. A bonded two-atom full-spinor check gives maximum adjoint and zero-step residuals of  $2.79 \times 10^{-14}$  and  $2.00 \times 10^{-13}$ , respectively.

An additional adaptive Z2Pack-driven stanene surface uses 42 CP2K line calculations and ends with 15 accepted transverse lines, all satisfying the line-position, movement, and gap checks, and again gives  $\nu = 1$ . This calculation chooses its sampling independently but uses the same CP2K Hamiltonian and overlap implementation. The main-text figure shows the native samples without interpolation or artificial separation of degenerate centers. It is regenerated by `python3 validation/topology/plot_wilson.py`, which requires NumPy and Matplotlib. Raw `.wilson` files, inputs, logs, comparison reports, generated comma-separated figure data, and checksums accompany the figure.

## S3 Native Chern and Cross-Geometry Extension

### S3.1 Source versions and independent analysis routes

The Chern and cross-geometry checks were completed on 18 September 2026 with a separate development executable. They are distinct from the retained 17 September neon/stanene calculations and do not reassign those earlier results to a newer implementation. Source patches, executable records, library versions, and independent comparison scripts accompany the validation archive. The numerical comparison tools require NumPy and SciPy; Z2Pack 2.2.1 is an optional independent analysis dependency.

Table S1: Topology analysis routes and runtime requirements. All native routes act on Gaussian-basis electronic states without constructing maximally localized Wannier orbitals.

| Route                         | Quantity and sampling                                                                           | Dependencies and scope                                                                            |
|-------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Native Wilson/ $\mathbb{Z}_2$ | Wilson spectra, Berry phases, and time-reversal half-plane $\mathbb{Z}_2$ .                     | CP2K native kernels, including second-variational SOC. No Z2Pack or Wannier90 runtime dependency. |
| Native Wilson/ $C_1$          | Determinant-phase winding on a full closed reciprocal surface, selected by <code>CHERN</code> . | CP2K native kernels. Fixed-rank isolated subspace, closure and refinement checks.                 |

<sup>1</sup>R.-w. Zhang et al., Room temperature quantum spin Hall insulator in ethynyl-derivative functionalized stanene films, *Sci. Rep.* **6**, 18879 (2016), <https://doi.org/10.1038/srep18879>.

| Route                          | Quantity and sampling                                                                                                            | Dependencies and scope                                                                                                                                     |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Native inversion analysis      | Even/odd irreps, Fu–Kane indices and inversion-subgroup EBR signatures at TRIM.                                                  | Separate native extension, Section S4. No general space-group catalogue.                                                                                   |
| Native parameter-mesh analysis | $C_1$ on two parameters and the second Chern character pairing $C_2$ on four parameters, using physical cross-geometry overlaps. | <code>topology_phasons</code> reads <code>STATE_EXPORT</code> files. One process with optional OpenMP. No Python, Z2Pack, or Wannier90 runtime dependency. |
| External Z2Pack route          | Independent loop/surface analysis of CP2K overlaps, with optional adaptive CP2K sampling.                                        | Python and Z2Pack. Optional analysis and cross-validation, not the implementation of the native invariants.                                                |

### S3.2 Parameter-mesh discretization and physical closure

Let  $Q_\mu(x)$  be the polar-unitary selected-state overlap directed from mesh vertex  $x$  to  $x + \hat{\mu}$ . All boundary links include the specified physical sewing. The oriented plaquette is

$$P_{\mu\nu}(x) = Q_\mu(x)Q_\nu(x + \hat{\mu})Q_\mu^\dagger(x + \hat{\nu})Q_\nu^\dagger(x). \quad (\text{S1})$$

With this link orientation, the two-parameter estimator is

$$C_1^{\text{mesh}} = -\frac{1}{2\pi} \sum_x \text{Arg det } P_{12}(x). \quad (\text{S2})$$

Its sign matches the determinant Wilson-phase winding convention of the main text. For four parameters, define the full principal matrix logarithm  $F_{\mu\nu}(x) = \log P_{\mu\nu}(x)$ . Each  $F_{\mu\nu}$  is anti-Hermitian and already includes the plaquette area. The estimator is

$$C_2^{\text{mesh}} = -\frac{1}{4\pi^2} \sum_x \text{Re Tr } [F_{12}F_{34} - F_{13}F_{24} + F_{14}F_{23}](x). \quad (\text{S3})$$

The full matrices are needed for a non-Abelian occupied subspace. Equation (S3) approximates the second Chern character pairing and is not generally the second Chern class. Its finite-mesh value is reported without integer rounding.

Physical links use  $C_a^\dagger O_{ab} C_b$ , with both spinor components summed when present. The cross-geometry AO matrix  $O_{ab}$  retains the moving contracted Gaussian centers, periodic images, and reciprocal boundary phases. The cell, atom count and selected rank remain fixed throughout the family. Endpoint checks use explicit atom permutations, lattice translations, basis matching, and selected-subspace sewing. The code reports AO-metric errors, sampled selected/excluded-band gaps, minimum link singular values, and principal plaquette phases. A refined mesh must preserve the same physical family. Sampled isolation and invariant convergence do not by themselves establish a bulk spectral gap between samples. An indirect gap is interpreted only for a lowest-band prefix with a justified common energy reference across geometries.

### S3.3 Retained validation results

The extension was rebuilt with GNU Fortran 16 and OpenBLAS. Its curvature and snapshot unit tests and the existing Wilson/ $C_1/\mathbb{Z}_2$  unit test passed with two OpenMP threads. All four inputs in `QS/regtest-topology` passed with two MPI ranks and two OpenMP threads per rank. The auxiliary parameter-mesh executable itself was tested as one process, not as an MPI-distributed postprocessor.

The native tests cover singular or underresolved links, metric and seam failures, determinant phases, non-Abelian gauge covariance, and model refinement. For the occupied doublet of the

four-dimensional Dirac model at mass  $-3$ , the  $8^4$  estimate is  $-0.7424128463$  and is closer to the continuum value  $-1$  than the  $6^4$  result. This is a finite-resolution regression check, not a claim of a converged integer at  $8^4$ . Independent Python/native comparisons additionally check gauge changes and reversal of parameter orientation.

For archived CP2K state exports covering skew cells, multiple centers, SOC spinors and reciprocal-boundary links, the largest difference between native and independently evaluated Gaussian overlaps is  $1.03 \times 10^{-15}$  with either one or two OpenMP threads. A cross-geometry overlap agrees to  $5.56 \times 10^{-16}$ . Table S2 reports physical atomic-translation pumps. The two-parameter case varies one reciprocal coordinate and one periodic atomic displacement. The separable four-parameter case uses the ordered coordinates  $(k_x, k_y, \phi_x, \phi_y)$ .

Table S2: Native physical Gaussian-snapshot tests. The minimum singular value is measured before polar unitarization. The phase column reports the largest diagnostic plaquette phase in radians. The  $4^4$  separable case tests the four-parameter physical workflow, not generic  $C_2$  convergence.

| Mesh           | Unrounded pairing     | Minimum singular value | Maximum phase |
|----------------|-----------------------|------------------------|---------------|
| $4 \times 4$   | $C_1 = -1.0000000000$ | 0.234278               | 0.402344      |
| $8 \times 8$   | $C_1 = -1.0000000000$ | 0.609933               | 0.099083      |
| $12 \times 12$ | $C_1 = -1.0000000000$ | 0.780474               | 0.044040      |
| $4^4$          | $C_2 = -1.0000000000$ | 0.233877               | 0.402347      |

The native  $4 \times 4$  to  $8 \times 8$  physical refinement check passed at an invariant tolerance of  $10^{-8}$ . The independent comparison-tool suite passed 34 tests without Z2Pack, with five optional-adaptor tests skipped, and all 39 tests with Z2Pack installed. The accompanying `validation/topology/native-phasons/` directory contains the machine-readable overlap and pump results, test logs, and source-version record. These are implementation tests and introduce no new material prediction or change to the neon/stanene results.

## S4 Native Inversion Representations and Symmetry Signatures

The inversion extension was validated separately from the phason extension and from the earlier Si/Al and Wilson calculations. Its source patch, base snapshot, inputs and outputs remain in `validation/topology/inversion-tqc/`. These records identify the original tests without retrospectively assigning them to another executable.

The WANNIER90 print section accepts KPOINTS\_SOURCE TRIM, SOC T, TIME\_REVERSAL T and INVERSION\_TQC T. TQC\_DIMENSION selects the four  $k_z = 0$  TRIM or all eight three-dimensional TRIM. PARITY\_ORIGIN specifies the inversion center in fractional direct coordinates. The occupied spinor bands must be a complete lowest-band prefix, with excluded states available for checking spectral separation. A PARITY-only mode also accepts isolated scalar subspaces. Neither route requires Z2Pack, Wannier90 execution or a Wannier fit. The TRIM calculation holds the converged SCF potential fixed and does not require the SCF integration mesh to contain those points. The two-dimensional option samples the fractional  $k_z = 0$  plane of a three-dimensionally periodic cell, not a separate two-dimensional Poisson boundary condition. The inversion center is specified explicitly rather than inferred from a space-group search.

For an orbital of angular momentum  $l$  on atom  $a$ , inversion about  $\mathbf{c}$  maps it to an identical Gaussian orbital on atom  $b$  when  $2\mathbf{c} - \mathbf{r}_a = \mathbf{r}_b + \mathbf{L}$  with integer  $\mathbf{L}$ . The coefficient multiplier is  $(-1)^l \exp(2\pi i \mathbf{k} \cdot \mathbf{L})$ . The code verifies atom-kind matching, involution, metric covariance,  $C^\dagger S C = I$ , unitarity and Hermiticity of the projected inversion matrix, and its commutator with the selected eigenvalues. Spinful time reversal is checked through the sewing matrix of  $\Theta = i\sigma_y K$ . Parity multiplicities must be even before division into Kramers pairs. The dimensionless and geometry tolerances are  $10^{-6}$  and  $10^{-6}$  bohr, respectively. The energy-commutator tolerance is

$10^{-6}$  hartree.

For  $N$  occupied Kramers pairs and odd counts  $n_{\kappa}^{-}$  at  $\mathbf{k} = \kappa/2$ ,  $\kappa \in \{0, 1\}^d$ , the parity index is  $\nu = \sum_{\kappa} n_{\kappa}^{-} \bmod 2$ . In three dimensions the weak indices restrict this sum to  $\kappa_i = 1$ , and the convention for the inversion indicator is  $z_4 = \sum_{\kappa} n_{\kappa}^{-} \bmod 4$ . The analytic inversion-subgroup EBRs are local even or odd Kramers pairs at centers  $\mathbf{c} + \mathbf{a}/2$ . Define

$$d_{\mathbf{a}} = 2^{-d} \sum_{\kappa} (-1)^{\mathbf{a} \cdot \kappa} (N - 2n_{\kappa}^{-}). \quad (\text{S4})$$

An integer atomic-signature decomposition exists when all  $d_{\mathbf{a}}$  are integers and  $N - \sum_{\mathbf{a}} d_{\mathbf{a}}$  is even. A nonnegative decomposition additionally requires  $N \geq \sum_{\mathbf{a}} |d_{\mathbf{a}}|$ . The implementation makes these decisions with integer arithmetic. Atomic compatibility concerns the symmetry signature, not a proof of trivial topology. Likewise, a signed-only decomposition does not by itself establish fragile topology. This inversion-only EBR test is distinct from the subsequent native little-group and explicit-segment analysis described below; it does not become a general EBR classifier when that additional analysis is used.

Table S3: Additional native parity checks with four occupied Kramers pairs. The four columns refer to  $(k_x, k_y) = (0, 0), (1/2, 0), (0, 1/2), (1/2, 1/2)$  at  $k_z = 0$ . Coordinates of the chosen inversion center are fractional.

| System  | $\mathbf{c}$      | $n_{\kappa}^{-}$ |   |   |   | $\nu$ | Atomic signature |
|---------|-------------------|------------------|---|---|---|-------|------------------|
| Neon    | $(1/2, 1/2, 1/2)$ | 3                | 3 | 3 | 3 | 0     | Compatible       |
| Stanene | $(0, 0, 0)$       | 1                | 2 | 2 | 2 | 1     | Obstructed       |

These tests retain the basis, cutoff, geometry and SCF meshes of the earlier material examples, replacing the Wilson paths by TRIM sampling. The three-dimensional neon test, with inversion center at the origin, also gives zero strong and weak indices and  $z_4 = 0$ . All seven inputs in the extended `QS/regtest-topology` set passed with two MPI ranks and two OpenMP threads per rank. The stanene parity result was independently reproduced with one rank and one thread. A single-rank, multiple-thread run exposed inconsistent state normalization in the existing local k-point diagonalization path for this build. The new metric checks reject that case. This extension does not resolve or validate that threaded path.

The lightweight CI fixture subsequently reduces the stanene SCF mesh from  $8 \times 8 \times 1$  to  $4 \times 4 \times 1$  and suppresses unused restart files. It still evaluates all four TRIM and reproduces the odd-pair counts  $(1, 2, 2, 2)$  and  $\nu = 1$ . This fixture is distinct from the archived  $8 \times 8 \times 1$  manuscript calculation and does not replace it or establish k-mesh convergence.

The native unit test covers multidimensional-character decomposition, nonorthogonal metrics, degenerate-state gauge changes, broken-symmetry rejection, all inversion-center EBRs and strong,  $z_4 = 2$  and signed-only examples. Independent enumeration of all two-Kramers-pair atomic sums matches the solver for all 81 two-dimensional and 6,561 three-dimensional candidate parity signatures. A deliberately incorrect inversion center is rejected in an end-to-end calculation. TRIM sampling does not prove a gap over the full Brillouin zone, and the SOC subspace still requires its usual convergence checks.

## S5 Native Little-Group and Compatibility Validation

The native algebra constructs unitary projective character tables, Wigner corepresentation restrictions, and endpoint-to-segment branching without a runtime dependency on Spgrep or a pretabulated irrep catalogue. The electronic-structure path uses SPGLIB to detect spatial operations and, when requested, adds their nonmagnetic time-reversal extension. It retains centering translations and tests the metric, band-subspace closure, energy commutators and the complete semilinear multiplication law before accepting band multiplicities.

The all-setting sweep covers all 530 Hall settings of the 230 ordinary space groups. At each setting, it samples  $(0, 0, 0)$ ,  $(1/2, 1/2, 1/2)$ ,  $(1/2, 0, 0)$ ,  $(1/3, 1/3, 0)$ ,  $(0, 0, 1/4)$  and  $(0.173, 0.217, 0.319)$  with scalar and spinful factors. The invariant cell metric is obtained by averaging over the supplied rotations. Both the ordinary group and its grey extension are checked. Generic-site orbits with physical spin multiplicity provide fixed-rank compatible band data. Segments run from Gamma to the other sampled points and from Gamma to its reciprocal image  $(0, 0, 1)$ . This tests directed-image monodromy as well as ordinary endpoint restrictions; it does not sample every possible line or material.

| Checked object                                        | Successful cases |
|-------------------------------------------------------|------------------|
| Ordinary/grey scalar/spinful factor systems           | 12,720           |
| Complete unitary projective character tables          | 6,360            |
| Endpoint tables with an antiunitary stabilizer        | 5,324            |
| Ordinary/grey directed segment comparisons            | 12,720           |
| Segments retaining an antiunitary stabilizer          | 4,462            |
| Gaussian directory assertions, serial / two MPI ranks | 69 / 69          |

Table S4: Native representation validation on 24 September 2026. The counts refer to algebraic cases or numerical assertions, not distinct material calculations.

Analytic cases separately test magnetic half translations, spin quaternion doublets, complex conjugate  $C_3$  pairs, twisted changes of factor gauge, and invalid pairing/doubling counts. Segment cases include displaced-origin PT pairs, endpoint-only time reversal, magnetic-glide Kramers planes, and spinful glide partner switching: a two-band hourglass connection is obstructed, whereas the complete four-band space is compatible. Unitary screw and mirror branching, reciprocal-image monodromy and changed coset representatives are also checked. The complete Hall-setting suite passes in release and with bounds checking and floating-point traps enabled.

The Gaussian directory uses UZH MOLOPT bases for its GPW fixtures and a contracted all-electron basis for GAPW. Ne tests exercise scalar and post-SCF GTH-SOC multiplets at Gamma, a generic PT-symmetric point and nonsymmorphic screw sectors. Scalar He/GPW, He/GAPW and Ne/SOC close a four-segment reciprocal loop; a separately selected single screw branch must report its monodromy obstruction. The generated Wilson-link test also exercises repeated refinement. Serial runs use two OpenMP threads; MPI runs use two ranks with two threads each and the thread-safe BLAS settings stated below. An earlier wider corepresentation integration pass additionally gave 110/110 serial and 111/111 MPI assertions across topology, Kubo, localizer and quadratic directories. These are separate test scopes, not numbers to add into a material benchmark total.

A separate local algebra sweep uses SPGLIB 2.7 magnetic data in the supplied setting for every one of the 1,651 magnetic UNI numbers. At the same six points and two spin choices, all 19,812 factor cases pass; 14,088 contain an antiunitary stabilizer, and 5,724 do not. For 10,154 primitive-cell tables supported by an independent Spgrep 0.7.0 comparison, dimensions, Wigner types and unitary restriction characters agree within  $7.11 \times 10^{-15}$ . Both calculations use the same supplied factor system, so this is not an independent validation of symmetry detection. Moreover, a separate semilinear-product check did not validate every explicit antiunitary matrix produced by that reference calculation. Those matrices are not imported into CP2K; the comparison is limited to types, dimensions and unitary restrictions. CP2K’s actual Gaussian sewing matrices pass their own full product check.

The separate native site-induction test samples fractional positions  $(0, 0, 0)$ ,  $(1/2, 0, 1/4)$  and  $(0.173, 0.217, 0.319)$  at the same six reciprocal points. Ordinary and grey groups, each with scalar and spinful factors, give 38,160 site/k-point cases over all 530 Hall settings. There are 67,824 local-representation columns evaluated and decomposed, and the same number of directed compatibility comparisons. Their atomic band rank equals the orbit multiplicity times the local irrep or corepresentation dimension. This construction preserves conventional-cell translation

sectors rather than silently converting them to a primitive-cell catalogue.

An independent check constructs Cartesian s/p/d orbital traces and, for spinors, multiplies them by the physical  $SU(2)$  trace. After matching the onsite orbital content, its explicitly translated orbit trace agrees with the corresponding induced character sum in 114,480 comparisons. The numerical acceptance tolerance is  $10^{-7}$  for these summed traces and decomposition checks; group and orbit geometry use  $10^{-9}$ . Analytic checks additionally cover inversion-center parities, screw orbits, spin-quaternion doublets, and complex local orbitals related by an antiunitary half translation. These are algorithmic reference tests, not new material calculations or a new production input mode. Additional scalar and spinful checks preserve the characters under passive origin shifts and unimodular changes to a sheared cell, and verify the expected Bloch rephasing under integer changes of operation representatives. The full database sweeps pass in serial, two-rank MPI, and a separate bounds-checked build with floating-point traps.

The native Wyckoff generator separately constructs 3,467 affine site-symmetry families across all 530 Hall settings and determines 5,648 directed specialization relations. External validation uses SPGLIB’s tabulated `Wyckoff.csv` coordinates, not the production fixed-set algorithm, to check dimensions, orbit multiplicities, family equivalence and closure containment. Every family matches bijectively, with largest affine residual  $2.22 \times 10^{-16}$ . The reference table’s version and checksum, comparison script and results are retained in the development record. No tabulated coordinates are read by the CP2K implementation.

For each generated seed the separate spinful induction kernel confirms its orbit and site stabilizer. Combined arbitrary origin shifts, nonorthogonal unimodular cell changes, integer shifts of operation representatives and reordered operations preserve the families and their specialization graph. Analytic tests cover inversion centers, intersecting mirrors, fixed-point-free screw axes, grey groups and an antiunitary half translation. Failure to resolve a generic stabilizer or an arithmetic/work limit returns no partial list. Maximal site groups are identified geometrically; this is not an elementarity test for their induced representations. The complete Wyckoff sweep passes in serial release, two-rank MPI and bounds-checked debug builds.

The subsequent atomic-signature module combines all generated sites and onsite irreps/corepresentations over the six reciprocal points listed above. Ordinary/grey and scalar/spinful variants yield 2,120 matrices, 31,079 provenance-preserving columns and 707,038 integer entries. Seven directed comparisons per column include the supplied point connections and a reciprocal-image loop, totalling 217,553 column/segment checks. Each block reproduces the column’s band rank; antiunitary rows count full corepresentations rather than treating all states as Kramers pairs. Additivity, permuted operation rows, and integer coset rephasing are checked. All sixteen inversion-center signatures and the generic-site signature are recovered analytically.

These full sweeps pass in serial release, two-rank MPI and debug with bounds and floating-point checks. A local optimized-compiler issue in automatic `MATMUL` result resizing was independently reproduced without CP2K and avoided by explicitly dimensioning the test vectors; no physical reference values were adjusted. The compiler reproducer and validation logs are retained separately from material results. No new general-classification input mode, elementarity assertion or complete reciprocal-stratum coverage follows from these matrix tests.

The integer-signature extension factors these same 2,120 matrices using checked 64-bit Smith reduction. Each factorization verifies its full integer certificate. One signed target per matrix is reconstructed, and the nonnegative search recovers every atomic column and a three-column sum per matrix: 33,199 witnesses in total. These deliberately constructed targets test recovery, not a classification of DFT bands.

An independent unit program compares Smith invariants with the gcds of all minors and computes the determinants of both unimodular transforms for 384 matrices with dimensions one through four. The rank-zero through rank-four case counts are 38, 167, 103, 60 and 16. The analytic inversion Fourier criterion supplies a separate exact oracle for every odd-parity-count pattern with two Kramers pairs: among 81 four-TRIM patterns, 41 admit signed and 33 admit nonnegative atomic signatures; among 6,561 eight-TRIM patterns, the counts are 241 and 129.

Thus signed-only cases are tested explicitly. For every  $2 \times 3$  matrix with entries in  $\{0, 1, 2\}$  and every two-component target in  $\{0, \dots, 4\}^2$ , a brute-force enumeration of coefficients zero through four supplies another oracle, covering 18,225 systems. Coefficients beyond four cannot contribute to a target in this range through any nonzero nonnegative column. Zero columns, deficient rank, invalid data, unsafe arithmetic and exhausted node budgets have distinct tests. The latter return an unresolved decision, not a symmetry obstruction.

The expanded matrix tests pass in optimized serial and two-rank MPI runs, and with the integer kernel compiled using bounds checks, floating-point traps and undefined-behavior instrumentation. MPI repeats the native algebra on each rank and does not distribute the catalogue construction. The analytic matrix path also passes with the test's SPGLIB preprocessor option disabled, linked against the existing library; this is not a full no-SPGLIB rebuild.

The Gaussian connection is enabled by `ATOMIC_SIGNATURES`. It compares only physically validated, sampled band subspaces with the generated reference matrix, preserving conventional-cell sectors. Band characters are rephased for integer changes of operation representatives, and the entire product/factor correspondence is checked, including antiunitary conjugation. All 2,120 algebra cases recover the same signature after operation and irrep permutations and such coset shifts. The shifted factors are regenerated by the separate little-group constructor rather than copied from the mapping formula. Inconsistent factors and unmatched operations are rejected. These expanded sweeps pass in optimized serial, two-rank MPI and with the new mapping kernel under bounds and undefined-behavior checks.

In the physical fixtures, scalar Ne at Gamma, Ne/SOC at Gamma and a generic point, centered/sheared He, and complete He/GPW, He/GAPW and Ne/SOC screw branches all have verified signed and nonnegative atomic signature witnesses. The independently obstructed single He screw branch has neither. The refined Wilson fixture repeats the comparison on both its four- and twelve-point sets. An orthorhombic He TRIM fixture sets `ATOMIC_SIGNATURE_MAX_NODES` to zero: signed membership is established, while nonnegative membership remains explicitly unresolved. This separates absence of a witness from proof of absence. The Gaussian-integration driver passed 73 assertions serially and with two MPI ranks, consisting of four native unit programs and 69 Gaussian-directory assertions. Existing physical reference values are unchanged; the new fixture uses the same UZH MOLOPT/GTH family as the other GPW tests. An independent reader then checks the exported records using arbitrary-size integer arithmetic: each of the serial and MPI runs contains 11 matrices and 19 signed/nonnegative witnesses satisfying  $Ax = b$  exactly, with consistent row/column band ranks. This verifies the stored certificates rather than independently classifying the negative cases. The reader and compressed output records accompany `validation/development/gaussian-signature-validation.md`.

The subsequent reciprocal-geometry test enumerates fixed families of  $(-1)^a W^{-T}$  for all 530 Hall settings and their grey extensions. Across these 1,060 geometric groups it retains 18,700 connected fixed-torus closures, 57,357 specialization incidences and 18,730 primitive integer tangent cycles. Full original cosets are restored when checking spinful stabilizers. At each setting, independent direct-transpose tests on reciprocal meshes with denominators 2, 3, 4 and 6 assign every point to exactly one matching family; this gives 333,900 point checks. Joint direct-cell shears, origin shifts, integer coset changes and operation permutations preserve the family and incidence correspondences. Analytic cases separately cover time reversal,  $PT$ , magnetic half translations, screw-invariant lines, centering and orthorhombic intersections.

For each ordinary/grey group, scalar and spinful atomic reference matrices are constructed at the automatically generated seeds and incidence images. Every atomic column obeys each directed restriction and primitive-cycle relation, giving 2,481,572 column/edge comparisons. This explicitly retains reciprocal shifts, nonsymmorphic phases and antiunitary restrictions. The complete sweep passes in serial and on both MPI ranks; the MPI run repeats the algebra, rather than distributing it. The new reciprocal module and its test also pass with bounds checks, floating-point traps and undefined-behavior instrumentation; their dependencies use the existing library. A database-free analytic test build passes against that library, not against a full no-

SPGLIB rebuild. The updated focused driver passes 74/74 serial and 74/74 MPI assertions: five native programs and the unchanged 69 Gaussian assertions. Logs and reproduction details are in `validation/development/reciprocal-strata-validation.md`. These are the geometric-stage results; fixed closures and their incidence/cycle graph are not a complete equivariant open-cell complex or a global connectivity classifier.

### S5.1 Automatic Gaussian symmetry sampling

The subsequent `KPOINTS_SOURCE SYMMETRY` integration generates and refines those paths directly at the frozen SCF Hamiltonian. Four Gaussian fixtures cover scalar He/GPW with UZH MOLOPT/GTH, all-electron He/GAPW with UZH TZVPP functions, Ne/GTH-SOC with its complete scalar AO space in the second variation, and a unimodularly sheared He cell with consistently transformed coordinates. They each yield 15 fixed families, 56 sampled points and 82 directed segments, all compatible. Each has signed and nonnegative atomic-signature witnesses. The focused driver now passes 106/106 assertions serially and with two MPI ranks: five native programs and 101 Gaussian assertions. The 32 added assertions do not alter the previous energy or symmetry references.

An independent file-level replay reconstructs NNKP inputs from the recorded fractional points and reciprocal-image edges. The rectangular NNKP neighbour format needs extra, nonduplicate padding connections; comparisons of segment data use the original directed edges only. Across the four cases, selected eigenvalues differ by at most  $2.1 \times 10^{-13}$  eV and band characters by at most  $2.2 \times 10^{-14}$ . Endpoint-to-line restrictions, the full atomic signature matrix and its integer witnesses agree exactly. The native path creates no `.mmn` file, whereas the replay exercises the existing overlap exporter. This comparison tests the new sampling integration against the established explicit-point path, not an independent electronic-structure implementation.

Increasing the number of points per original segment from three to five gives 138 points and 164 compatible segments for He/GPW. Removing time reversal retains valid unitary restrictions and reports zero antiunitary segments. Negative tests reject a too-small point budget, an invalid subdivision count and a conflicting Wilson-loop request. Selecting one of the two He screw branches fails the isolation check at a generated degeneracy instead of yielding an atomic conclusion. Analytic subdivision tests independently check lifted endpoints, uniform increments and total winding for positive, negative and multiple boundary crossings, including budget failure without mutation. The full 1,060-group geometry sweep is repeated with path subdivision. Records and the replay procedure are retained in `validation/development/automatic-sampling-validation.md`.

### S5.2 Joint compatibility lattice and independent certificates

The following baseline precedes the star-identification extension described in the next subsection; its archived certificates are retained. The next extension assembles all endpoint-to-line restrictions into one integer matrix  $C$ , expressed in the same irrep/corepresentation coordinates as the atomic matrix  $A$ . In particular, a reciprocal-image self-edge subtracts two restrictions in the same point block and retains nonsymmorphic monodromy. An exact Smith kernel  $K$  and integer atomic coordinates  $Y$  satisfy  $CK = 0$  and  $KY = A$ . The Smith form of  $Y$  supplies the free and finite factors of  $\ker_{\mathbb{Z}} C / \text{im}_{\mathbb{Z}} A$ .

The 530-setting ordinary/grey and scalar/spinful sweep verifies every generated atomic column in the joint kernel with zero quotient class. Small-group variants reverse all directed edges and reciprocal shifts, reject a deliberately incompatible atomic column and enforce a tiny memory budget without returning partial data. Separate integer tests include a nonsaturated constraint row, a free-plus-torsion quotient, zero-dimensional kernels and incompatible targets. The four/eight-TRIM inversion matrices with equal-rank constraints yield  $\mathbb{Z}_2$  and  $\mathbb{Z}_2^3 \times \mathbb{Z}_4$ , respectively. For all 81 and 6,561 two-pair parity patterns, zero class is equivalent to the independently tested signed atomic membership.

The full sweep passes in optimized serial, on both MPI ranks and with the new integer/compatibility and reciprocal kernels compiled with bounds, floating-point traps and undefined-behavior checks. All printed graph ranks/factors agree. The instrumented database-free analytic subset also passes against the existing library, not a full no-SPGLIB build. MPI repeats the algebra independently on each rank.

An independent SymPy 1.14.0 reader validates eleven exported Gaussian quotient records from each of the serial and two-rank MPI runs. It checks  $CK = 0$ ,  $KY = A$  and  $CA = 0$ , computes the rational rank of  $C$  and an independent integer Smith form of  $K$  to certify saturation, and factors  $Y$  independently. It also checks unimodularity of the exported quotient row map. Dividing its nonzero transformed rows by their divisors must give a primitive full-row-rank lattice; free rows must vanish. Finally it solves for each compatible target in  $K$  and recomputes the exported residue vector using arbitrary-size integers. Incompatible targets must not have a quotient-class record.

For the automatic He/GPW, He/GAPW and sheared-He fixtures,  $C$  has 109 retained equations and 91 coordinates. Its kernel has rank six, no free quotient factor and one  $\mathbb{Z}_2$  factor. The Ne/SOC fixture has 81 equations, 67 coordinates, kernel rank five and quotient  $\mathbb{Z}_2^2 \times \mathbb{Z}_4$ . All four physical targets have zero class. These are deliberately small atomic examples, not nontrivial topological-material discoveries. The explicit four-point complete screw fixtures have trivial quotients. The selected single screw branch violates compatibility and is assigned no class. By contrast, the sparse explicit He and refined Wilson graphs have free ranks four and 23. Those ranks expose missing constraints or identifications in the supplied graph and are not labelled physical topological invariants. Even a zero free rank does not establish completeness of omitted congruence constraints or BZ-wide isolation.

The official focused and exporter drivers together pass 165/165 assertions in each of SSMP and two-rank MPI. This comprises 129 native and little-group assertions and 36 existing export/k-point assertions; 23 new checks inspect compatibility, kernel rank, free/cyclic factors and actual target classes without changing existing physical reference values. Reproduction commands, the independent reader and compressed certificates accompany `validation/development/compatibility-lattice-validation.md`.

### S5.3 Star transport and published quotient comparison

The subsequent extension includes star-arm identifications in the same integer matrix, using the star-character transport formula derived in the main text. A threefold screw example checks genuinely complex characters, antiunitary conjugation, reciprocal shifts and independent integer changes of destination representatives. A star-only graph reduces nine local coordinates to three independent ones without requiring any line edges; a deliberately mismatched target is rejected. Corrupted factors, nonfinite characters and unsafe phases must not return a permutation. Every retained star link in the complete Hall-setting sweep agrees with independently induced atomic columns; inverse maps and all alternative connecting operations give the same full-corepresentation identification.

An independent development reader compares the resulting ranks and ordered nontrivial Smith factors with Tables I–IV and IX–XII of [Po, Vishwanath and Watanabe \(2017\)](#). It validates all 230 entries in each rank table and uses the SPGLIB Hall-to-space-group metadata solely to associate settings with the reference rows. All 2,120 ordinary/grey and scalar/spinful Hall-setting combinations agree, with no free factors on these automatically generated graphs. These are representation group comparisons, not 2,120 DFT calculations. They do not independently prove completeness of a user-supplied graph or the elementary nature of every generated atomic representation.

The Gaussian certificate reader also checks each exported star’s reciprocal geometry. The explicit He/GPW and He/GAPW screw examples now contain ten equations for nine coordinates, still with compatible rank three and trivial quotient. The Ne/SOC screw has five equations, five

coordinates and rank two. The refined He Wilson graph contains 76 equations and 75 coordinates: its kernel rank falls from 42 to 23 and its free quotient rank from 23 to four. The partial two-point He graph still has four free factors; the single-branch screw obstruction remains incompatible. No energy, character or physical reference value is adjusted to obtain these results. The automatic Gaussian fixtures retain the ranks and finite factors of the baseline.

Reproduction scripts, reference-comparison records and the new certificates are archived separately in `validation/development/star-transport-validation.md`. The star-transport stage passed 170/170 assertions in each of SSMP and two-rank MPI: 134 native/little-group checks and 36 existing export/k-point checks. Five new Gaussian assertions count star links, including the zero-link monodromy control and the six links of the refined Wilson graph. The full star-aware sweep output is identical between optimized SSMP, each MPI rank and the instrumented build. The database-free analytic variant passes as well, using the existing library rather than a full no-SPGLIB build.

These algebra sweeps do not implement magnetic-space-group detection, a magnetic non-collinear ground state, canonical irrep labels, a complete Brillouin-zone connectivity graph or a full elementary-band catalogue. Explicit-segment compatibility is necessary, not sufficient, for a gapped band subspace. The retained summaries and test logs are in `validation/development/`; no production material classification is inferred solely from a successful database sweep.

#### S5.4 Real-space induction and composite witnesses

The subsequent site-specialization extension evaluates finite site-group induction at lifted endpoints of the generated Wyckoff families. All 22,592 relations in the ordinary/grey scalar/spinful 530-setting sweep pass their independently generated band-character identities at eight reciprocal points, giving 180,736 checks. Transitive singleton equivalences and direct-sum witnesses also preserve every tested character and physical band dimension. The optimized SSMP, both MPI ranks and bounds/undefined-behavior instrumented executable produce identical 4,774-line result records. Intermediate-position tests cover inversion and mirror families, cell shear, shifted origin and integer coset changes. A separate analytic build omits the database interface; this uses the existing CP2K library rather than constituting a complete no-SPGLIB build.

The detected exceptional-space-group coverage matches Tables II–IV of [Cano et al. \(2018\)](#) in all 2,120 variants. A second comparison uses the independently archived Bilbao EBR data in `Crystalline.jl`. Every per-setting generator count agrees; counting each space group once gives 3,383 scalar and 2,258 spinful generators without time reversal, and 3,141 scalar and 1,616 spinful generators with time reversal. The archived dataset’s 2,258 differs from the aggregate 2,263 printed in the 2018 article. The comparison retains the dataset’s values and does not adjust a native count to obtain agreement. Neither these totals nor agreement of exceptional-space-group coverage establishes a canonical irrep-by-irrep identification.

For each of 15 Gaussian detail files per build, an independent arbitrary-precision reader verifies  $AE = A$ , the conserved band dimensions, the lifted endpoint geometry, every source stabilizer’s action on the entire displacement, singleton equivalence classes and recursive decomposition provenance. The physical fixtures include GPW, all-electron GAPW, post-SCF GTH SOC, sheared cells, screw sectors, TRIM and refined Wilson sampling. The official serial and two-rank MPI drivers each pass 171/171 assertions: six native unit programs, 129 Gaussian little-group assertions and 36 existing export/k-point assertions. The bounded new unit includes the spinless time-reversal exception at an  $S_4$  site in space group 84.

The reproducible comparison scripts, versioned reference hashes, native logs and Gaussian certificates are retained in `validation/development/site-induction-validation.md`. The reference tables are test oracles, not runtime dependencies or imported CP2K classification tables. These tests prove the returned decompositions and their integration, not a globally gapped material classification, canonical EBR labels or completeness of arbitrary reciprocal sampling.

## S5.5 Affine-reference and site-resolved EBR checks

The next reference layer uses the separately archived space-group generators and parametrized Wyckoff coordinates of Crystalline.jl. The test fixture closes the generators with exact rational arithmetic; the native reference path does not obtain operations from SPGLIB. All 1,731 named affine families match the generated closures, including their whole tangent spaces, dimensions and conventional-cell multiplicities. A second pass simultaneously changes the basis, origin, integer coset representatives and operation order and preserves every named specialization edge. The four ordinary/grey scalar/spinful variants give 6,924 affine correspondences.

After the conventional-cell centering factor is removed, the band dimensions of the 10,398 unreduced maximal-site columns agree with the independent EBR multisets separately at each Wyckoff position. No reference column is selected or discarded using the legacy `Decomposable` field. This test detects incorrect distributions among sites or band dimensions that total counts would miss, but does not assign canonical labels to equal-dimensional irreps. Optimized serial and two-rank MPI output is compared with a separately instrumented reference executable compiled without its SPGLIB database interface. The latter still links the existing CP2K library, and is not a complete no-SPGLIB CP2K build. Numerical guards additionally reject wrong tangents, repeated reference entries, incorrect multiplicities and nonfinite or unbounded coordinates. Reproduction scripts, source hashes and logs are retained in `validation/development/affine-reference-validation.md`.

## S5.6 Complete character correspondence with reference tables

The `irreptables` 3.1.0 package supplies independent single- and double-valued characters at the tabulated reciprocal points of all 230 space groups. The fixture reader uses the data files only, not IrRep’s character-generation implementation. It records a SHA-256 hash for each of the 460 source files. Centering is removed by an explicit primitive basis  $P$ , with  $W' = P^{-1}WP$ ,  $\tau' = P^{-1}\tau$  and  $k_{\text{native}} = -P^T k_{\text{table}}$ . The sign converts the table’s translation-character convention to the native  $\exp(-2\pi i k \cdot T)$  convention; the original character columns and spin lifts are retained. Without this conversion, central-projector audits reject 25 boundary-point tables. With the declared conversion, all 2,700 tables pass the independent audit. This is a convention check, not evidence of an error in IrRep.

Crystallographic coordinates printed to five decimal places are recovered as rational values with denominators at most 48 and changes strictly below  $5.1 \times 10^{-6}$ . Characters are not rationalized or rounded. A Cartesian intertwiner aligns the reference spin matrices with the integer rotations; unitary polar factors are used only to identify this coordinate frame, while comparisons retain the raw reference matrices. Native spin representatives are returned by the same factor-system construction that generated their signs. An independent recomputation after a different floating-point cell normalization is not used to infer those signs.

Every character column is compared, including distinct irreps with the same dimension. All 8,907 columns match; the largest residual is  $1.20921 \times 10^{-5}$ . The native construction tolerance remains  $10^{-9}$ ; the external comparison uses  $2 \times 10^{-4}$  to accommodate the printed reference precision. The 11,607 normalized case and label records are identical in optimized SSMP, separately on both MPI processes, and in a focused GNU Fortran build with bounds checking, floating-point traps and undefined-behavior instrumentation. The instrumented executable links the existing CP2K library; this is not a fully independent no-SPGLIB build or a multi-node scaling test.

The bounded native test covers ordinary and projective  $C_3$ , complex rephasings, label-permuting one-dimensional twists, interleaved antiunitary operations, and paired/doubled Wigner restrictions. An analytic body-centered three-screw boundary case also changes the Cartesian orientation, operation order and cell scale from  $10^{-100}$  to  $10^{100}$ . Inconsistent factor-gauge maps, repeated character columns, NaNs, unbounded values, and orthogonal dimension-preserving mixtures that are not representation characters are rejected. Full onsite EBR naming, arbitrary-cell standardization for named labels, and global connectivity are not implied by these reference correspondences. The

official SSMP and two-rank MPI drivers each pass 173/173 assertions: eight native unit programs, 129 Gaussian little-group checks and 36 existing topology/export/k-point assertions. No physical reference value was changed for this work. Scripts, provenance and numerical logs are retained in `validation/development/character-reference-validation.md`.

## S6 Bloch Kubo and Property-Symmetry Validation

The retained development report is `validation/development/bloch-response-summary.md`. It records distinct finite-cutoff regression fixtures and the higher-cutoff primitive/supercell comparison. For Ne with post-SCF GTH SOC and PBE at 2000 Ry, the  $3 \times 1 \times 2$  primitive mesh gives  $-34.923548219306689$  hartree, compared with  $-209.541289276241002$  hartree for six cells sampled explicitly at Gamma. Both print  $6.21105320 \times 10^{-9}$  S. At the inexpensive 200 Ry PADE regression settings, the respective conductivities are  $6.42389197 \times 10^{-9}$  and  $6.35401974 \times 10^{-9}$  S. Those distinct references are retained: the mismatch is not removed by changing a reference value.

Ne/SOC on a  $2^3$  property mesh uses four representatives. Bi<sub>2</sub>/SOC on  $3^3$  uses six; its printed nine-component tensor differs from full-grid by at most  $2.71 \times 10^{-15}$  relative to its largest entry. Primitive diamond/SOC on  $2^3$  uses three representatives with either K290 or SPGLIB. Its tensor difference is  $2.21 \times 10^{-11}$  in the same normalization. Metric, Hamiltonian and current reconstruction residuals are approximately  $2.1 \times 10^{-13}$ ,  $6 \times 10^{-11}$  and  $1.3 \times 10^{-11}$ , respectively. These are operator-covariance fixtures, not converged material transport predictions.

The final recorded directory drivers passed 132 serial and 162 MPI assertions. The MPI layout was four ranks with two OpenMP threads, and the serial build also used two OpenMP threads. Coverage includes moments, k-point/Wannier export, topology, Kubo, localizer, and quadratic directories; the MPI build additionally tests MUMPS and Tacho. Mathematical unit tests cover the constituent operator identities separately.

The Apple ARM runtime required one BLAS thread, a consistent thread-safe OpenBLAS preload, and an OpenMP stack of 128 MiB. Without these settings, failures occurred before the property code. An implicit-Gamma high-cutoff conversion also exposed a slow Miller-index lookup; the completed comparison uses an explicit Gamma input. These runtime qualifications must accompany the results, rather than be presented as production scalability.

## S7 Localizer and Quadratic Validation Records

The transferred developer notes in `validation/development/` document finite and periodic formulations, dense and sparse factorization, and their limitations. They are implementation records, not independent material benchmarks. The matrix tests include nontrivial and trivial Chern models, spin-mixed Kramers-pair models, origin and energy shifts, nonorthogonal congruences, invalid metrics, and gap closings.

Dense LDL tests cover both  $1 \times 1$  and  $2 \times 2$  blocks. Pfaffian tests independently cover odd permutations and recursive small-matrix reference polynomials. Sparse gap brackets are compared with generalized dense spectra, not with pivot sizes. Realification duplicates the spectrum and is accounted for in every inertia comparison.

For a larger molecular quadratic check, Bi<sub>2</sub> uses TZVP-MOLOPT-GGA-GTH-q5, GTH SOC, 68 spinor coefficients, four requested states and a 32-dimensional trial space. GPW and GAPW dense/iterative comparisons on four MPI ranks agree in gaps within  $6 \times 10^{-15}$  hartree, with all four metric-subspace singular values within  $10^{-10}$  of unity. This is separate from the smaller mandatory regression fixtures.

## S8 Submission Validation Still Required

The current record does not establish a nontrivial, size-converged DFT localizer classification for a large SOC flake, multi-node sparse scaling, or general 230-space-group TQC. The final archive should contain complete material inputs and potentials, basis and cutoff studies, spatial/energy/scale scans, independent band references, and timed runs with whole-process memory. Development diagnostics and planned work must not be merged into tables of completed material results.